Exploring the interaction between compact binaries and their host galaxy

Márk Jánosdeák

Student ID: 10748961

May 2024

The project was done under the supervision of Prof. Rene Breton and Dr. Soheb Mandhai

University of Manchester
Department of Physics and Astronomy

Abstract

In this MPhys thesis we investigate the behaviour of stellar systems receiving a natal kick in the Milky Way, using a computer simulation developed during the previous semester. The total retention fraction of stellar systems as a function of kick velocity is obtained and analysed for possible trends. An investigation is conducted into the kick velocity needed at different radii to reach 50% retention. Using a model for the total retention fraction, together with an observed velocity distribution for pulsar kicks an intrinsic kick velocity distribution is inferred. Moreover, a method for determining a distribution of possible radial birth locations of pulsars with known proper motions and distances is developed.

1 Introduction

The multiplicity frequency in stellar astrophysics refers to the probability of a star existing within a binary or multiple star system. Research indicates that Population I stars with masses exceeding 8 solar masses are more likely to be part of a binary or multiple star system (Duchêne & Kraus, 2013). The evolution of such systems can lead to the formation of compact binary systems, defined as systems containing at least one compact object (remnant of a dead star), such as a neutron star (NS), black hole (BH), or white dwarf (WD) (Tauris & van den Heuvel, 2006). Other types of compact binary systems include those where both companions are compact objects, such as neutron star - neutron star (NSNS) and black hole - neutron star (BHNS) pairs. The orbits of these systems decay until their merger, this together with the dissipation of distortions arising in the shape of the final black hole (coalescence), results in the emission of gravitational waves (Abadie et al., 2011). Coalescence involving a neutron star can produce electromagnetic phenomena like short-duration gamma-ray bursts and a kilonova (Li & Paczyński, 1998), the latter resulted from the radioactive decay of heavy elements formed during rapid neutron capture events (r-process). The former phenomenon originates from the disrupted material during the merger (Kouveliotou et al., 1993).

The detection of these merger events has become possible due to the first ground-based gravitational wave observatories (LIGO, VIRGO) (Abbott et al., 2016). Future space-based gravitational wave observatories like the Laser Interferometer Space Antenna (LISA) are expected to detect millions of compact binaries in our galaxy (Kremer & Breivik, 2017). This underscores the necessity for simulations of compact binary systems to explore potential trends surrounding them and their progenitors. The primary challenge of studying compact binary systems in our Galaxy lies in the current limitations imposed by the sensitivity of existing gravitation wave detectors. Our simulations will provide us with an insight into the spatial distribution of binaries within the Milky Way, which can be thoroughly tested against findings made by future space-based gravitational wave detectors. However, we can learn more about compact systems using objects that are possible to detect in larger numbers in our Galaxy, using electromagnetic methods, such as pulsars.

The evolutionary pathway of a massive star is a widely understood process (Lamers & M. Levesque, 2017), resulting mainly in a compact object, contingent upon the original star's mass. Stars with masses from the smallest to mid-range, in the range of $> 8\mathrm{M}_\odot$, leave behind a WD. Progenitors heavier than this threshold mostly undergo a core collapse supernova, leaving a NS or a BH in their wake. In Type II supernovae, which predominantly yield NS, a phenomenon known as a natal kick can occur. Natal kicks represent the velocity acquired by the NS relative to the rest frame of its progenitor system. Typically, for isolated pulsars and NS, the kick's magnitude falls within the range of 200-500 kmh^{-1} (Igoshev & Verbunt, 2018; Igoshev, 2020).

In this thesis, we are investigating the behavior of stellar systems experiencing their first natal kick, with a key focus on the examination of how these systems evolve over time under varying magnitudes of kicks. Since we are interested in the overall trends produced by a single kick to study the distribution of compact objects in the Galaxy, we ignore the internal configuration of the systems. Hence, in the following a star, binary and a stellar system or object are all used as equivalent phrases when discussing our work. To conduct the work outlined above, we developed a "toy-model" simulation inspired by the research conducted in Mandhai et al. (2022).

In the following, Section 2 outlines the work carried out during the previous semester, including the nature of assumptions made, along with links to the theory that gives rise to them. In Section 3, the method of simulation is discussed, together with the different investigations conducted using the simulated data and their results. Section 4 discusses the results and their implications, consequently giving suggestions for further investigations and improvements. Finally, in Section 5, a summary of the main results and conclusions are given.

2 Summary of the previous semester

During the previous semester, our main goal was to build up a computational simulation of the Milky Way, together with a random sampling method to generate stellar objects based on the baryonic mass distribution of the Galaxy. Furthermore, these stellar systems were evolved for a time period comparable to the Hubble time, while undergoing a supernova explosion resulting in a natal kick. We concluded this work by analysing individual orbits and outlining their characteristics. Here we provide a brief overview of the important assumptions made when developing the model, to support the understanding of the discussions further below.

2.1 Overview of the simulation

The Milky Way was simulated using analytical functions describing the gravitational potential and the density driving it. The potential profiles used were the following: the Navarro-Frenk White (Navarro et al., 1996) profile for the dark matter halo, the Miyamoto-Nagai (Miyamoto & Nagai, 1975) profile for the baryonic matter in the disk, and a spherical power-law density profile with exponential cutoff (Bovy & Rix, 2013) for the bulge. The functional description above was chosen after consideration of the existing models and the current observations of the rotation curve of the Milky Way. The choice to model the Galaxy using analytical potential profiles was taken considering computational efficiency. Each of the expressions have free parameters that have to be set and an overall scale factor in front of them when combining them. These variable parameters and the scale factors are based on the values used in Bovy & Rix (2013).

The goal of the random sampling method mentioned above was to create a way to generate random stellar systems with a distance distribution corresponding to their most probable

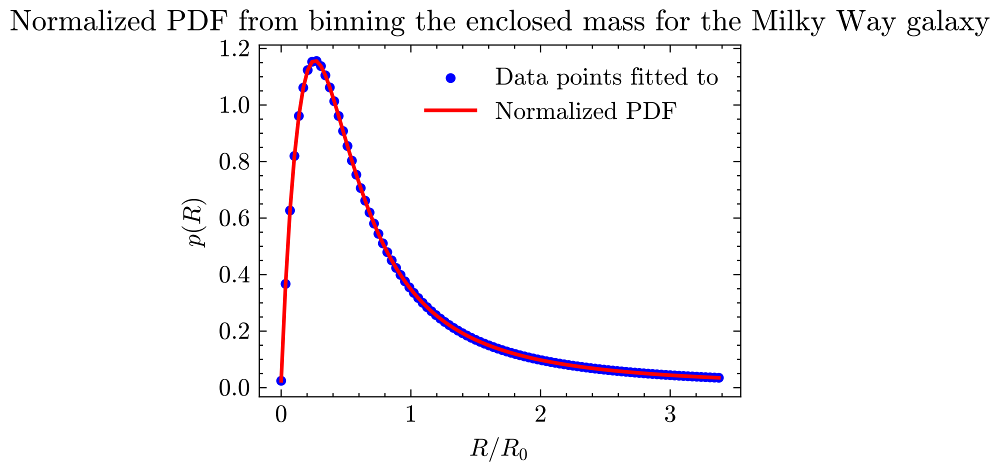


Figure 1: The figure above shows the PDF of the baryonic mass inside the Milky Way. It was created by binning the enclosed baryonic mass, to create a baryonic mass per shell data. Then a spline fit to these data points was created, hence creating a PDF. R_0 is defined to be 8 kpc.

birth locations. To infer an approximate birth distribution of early stars in the Milky Way to first order, we have made two assumptions, both of which arises from some experimental observations. The first is that the Milky Way has been relatively static over time (Yang et al., 2023) and the second is that the baryonic mass of the Galaxy is to a first order equal to the combined mass of the bulge and the disk (Hammer et al., 2007). Combining these two we arrived at the conclusion that the birth distribution of stars inside the Milky Way can be approximated by the current baryonic matter distribution of the Galaxy. Subsequently, this distribution was derived from the enclosed mass arising from the potentials and a spline function was fitted between the previously defined limits of the Galaxy, which are explained later, to create a probability density function (PDF) for the random sampling of stellar objects. The normalized PDF can bee seen in Figure 1.

When generating the stellar systems form the PDF discussed above, it was assumed that all stars are born in the Galactic plane ($z = 0$) and that all of them initially revolve around the Galactic center in circular orbits. The former is a well known approximation regarding pulsars used in various literature (Hobbs et al., 2005; Lyne et al., 1985; Deshpande et al., 1995) and the latter is justified by the fact that the relative difference between the velocity required to model realistic, slightly out of plane orbit and the additionally gained kick velocity is minuscule. Since, the potential profiles used for the description of the Milky Way are axisymmetric the azimuth angle of the stellar objects is redundant, hence initially it was set for a fixed value chosen to be zero.

The last crucial thing during the previous semester was to model the additional velocity received by the star systems, when undergoing a supernova explosion. This velocity gain was modelled by an isotropic directional fixed magnitude velocity kick, since there has not

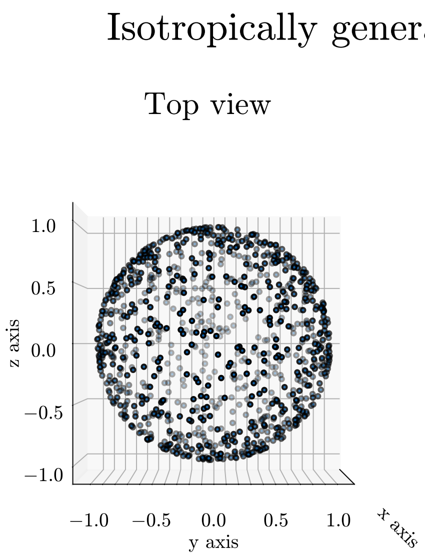


Figure 2: 3D plot representing the isotropicly generated unitary vectors on a surface of a unit sphere, using the Cartesian axes for reference. The top and the side view is given to illustrate the uniformity of points on the surface. Note, that here the Cartesian axes labels are not following the usual naming convention.

yet been significant evidence found for the preferred directions of the natal kicks (Bray & Eldridge, 2018).

2.2 Improvement on isotropic kicks

During the previous semester, when simulating the isotropic kick vectors, we used the following procedure. We drew a uniform random number from the interval $[-1, 1]$ for each Cartesian direction axis and re-normalized them for a unit vector on a sphere. However, it was later realized that this was a faulty method, since it produced over-density of points around the regions of the sphere that got projected down to from the area around the edges of the cube. To fix this we implemented a well known solution to the problem. We generated Gaussian random numbers in all three Cartesian directions and re-normalized them to unit length, hence giving us uniformly distributed points on the surface of a sphere. The results are presented in Figure 2.

3 Methods and Results

During this semester, we ran our simulation for different sample sizes of stellar systems and applied kick velocities. To begin with we investigated fixed magnitude kicks, such that all of the generated stars in the sample received the same magnitude kick, but in an arbitrary isotropically distributed direction. Initially we chose to investigate the magnitude range of

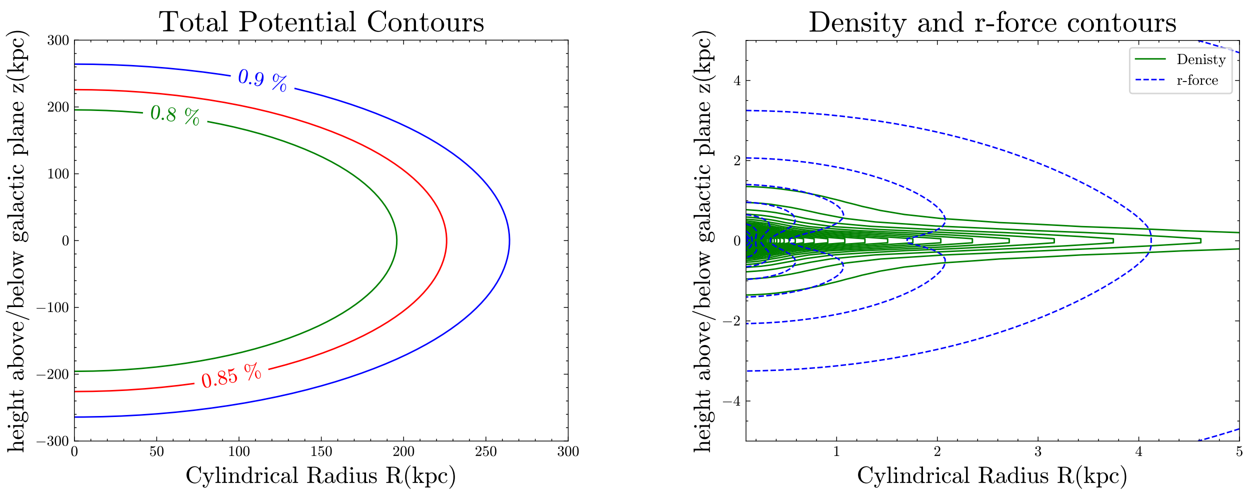
$50 - 250$ kms^{-1} in 50 kms^{-1} increments, since we are mostly interested in the population of pulsars that get retained, to carry out detailed analysis with the help of their orbital information. The choice of sample sizes simulated for was arbitrary and as such the powers of ten were chosen up to the fourth one. Although, from the data of the $10,000$ stellar object simulation we gained sufficient information we kept the 100 and $1,000$ sample sizes data as well to gain an insight into the size of fluctuations that appear due to the smaller sample size.

Before running the simulations we needed to decide the number of time-frames used when dividing the time interval (Hubble time) that we integrate the orbits for. From insights gained during the first semester in terms of the computational time and the resolution of the final data points we decided to use a 1,000 steps. These divide the Hubble time of 14 Gyr to 14 Myr segments equally spaced from each other. We point out here that the 14 Myr resolution for orbits near the Galactic Center (well within the 1 kpc radius) is not sufficient to fully resolve the characteristics of the systems here. However, in our investigations we are interested in the overall statistics of the orbital variables of these systems, thus this resolution is sufficient.

3.1 Placing a gravitational boundary on the Milky Way

We ran the simulations for samples up to $10,000$ and recorded all the orbital data. A method of classifying the simulated orbits needed to be developed. To achieve this, we defined a condition for the bound-unbound limit of an orbit within the Galaxy. Firstly, we tried to use equipotential contours of our total Milky Way potential, and define a bound level where 90% of the gravitational potential value at infinity lays. Due to the massive contribution of the dark matter halo to our Galaxy's mass and hence potential, this value lies hundreds of kiloparsecs away from the Galactic center. However, when talking about pulsars, most of the observed ones are located near the visible Galaxy (Manchester et al., 2005), hence if we use the above defined limits, all of the pulsars observed would be classified as bound. On the other hand, this evaluation provides certainly the most realistic gravitational bound-unbound limit. Since, stars belonging to the Milky Way have been discovered beyond 200 kpc (Bochanski et al., 2014). The plot of the contours of the total simulated Milky Way potential is presented in Figure 3a.

However, we opted for a different view, where the objects remaining within the visible regions of the Galaxy are to be considered bound and the ones escaping or orbiting at large distances are considered unbound; where a large distance is in the order of a few tens of kiloparsecs. Trying to find a different description for the bound-unbound limit of our simulated galaxy, we investigated the density contours intrinsic to the potential and the restoring force arising from the potential. These are illustrated in Figure 3b, to show the corresponding structural differences. We decided against the density contours since the shape features are only significant when looking at the close regions of the Galactic center and they



(a) Dark matter contours

(b) Density and restoring force contours

Figure 3: In the left figure, the contours of the total simulated Milky Way potential, including the dark matter contribution in the R-z plane is visible. The contours are the 80, 85 and 90 percent of the value at infinity. In the right figure, the contours for the matter density giving rise to our potentials and the restoring force coming from the potentials itself in the R-z plane is presented.

roughly tend to the restoring force contours at larger radii, and due to the computational difficulties surrounding them. The restoring force contours were promising, since they tended to the shape of the dark matter contours at larger radii, hence decided to keep them as strong candidates.

Thirdly, we used the condition defined during the previous semester's work (Jánosdeák, 2024), which put the boundary of the Milky Way at 27 kpc (R_G). Namely, where 90% of baryonic matter is included in a sphere of radius equal to the boundary distance. However, this is more of a baryonic limit and does not include the dark matter halo of the Galaxy, which still provides gravitational attraction. We decided to set the gravitational potential in the galactic plane ($z = 0$) at R_G to be the minimum, and we drew an equipotential contour in the R-z plane with the minimum value to outline the limits of our Galaxy. The result is presented in Figure 4.

This side view of our Galaxy does coincide well with the visible matter of spiral galaxies similar to ours. Since, the gravitational potential at low radii is very large due to the masses of the bulge and the disk. This is clearly visible in Figure 1, produced in the previous semester (Jánosdeák, 2024). Finally, we decided next to our last description, using the hard boundary of the baryonic potential at R_G. We also compared it to the restoring force contour at R_G and found that those two are almost identical, so it was up to preference which one we choose and our decision was to go with the potential one, because it is computationally more beneficial when using `galpy`.

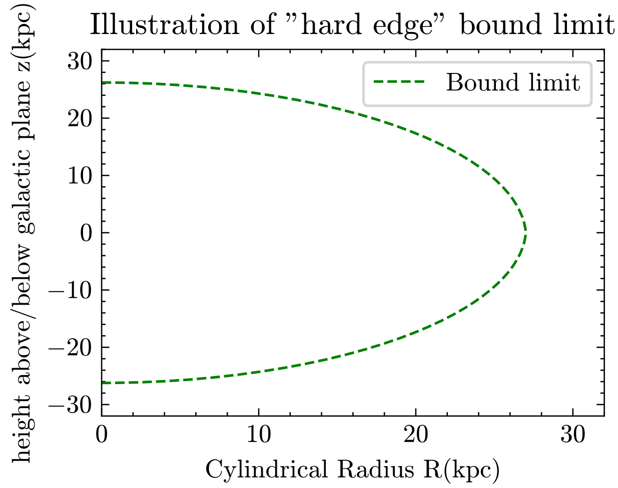


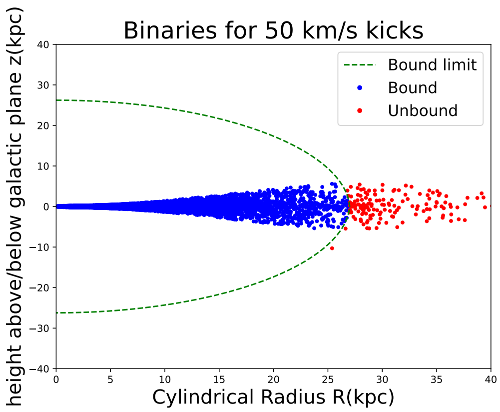
Figure 4: Illustration of the bound limit using the hard boundary definition of the Milky Way at R_G. The R-z plane is used for illustration with both of the axes having units of kiloparsecs.

3.2 Distribution of simulated objects

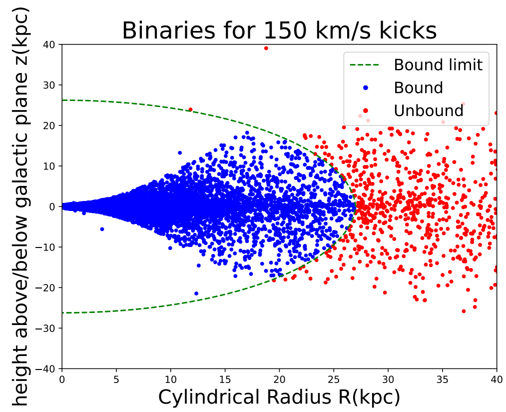
After determining the bound-unbound condition, we analysed our simulated data. To visualise the orbits one point needed to be selected for each, that was not a direct task. Due to the analytical nature of our potentials and the fact that orbits within our simulation are solely influenced by the gravity of a single galaxy. After careful consideration we found that the most sufficient way to show our results would be to plot the point on each stellar objects path that is the least gravitationally bound. To achieve this, we calculated the gravitational potential at the position of each time step within the orbit's integration and recorded it. We plotted this in the R-z plane together with the indication of the bound-unbound limit. In Figure 5 a selection of constant kick magnitudes are plotted that illustrate the distribution of the objects with growing kick magnitudes. The objects termed bound have been colour coded with blue and correspondingly the ones termed unbound red.

3.3 Retention fraction and partial retention fraction

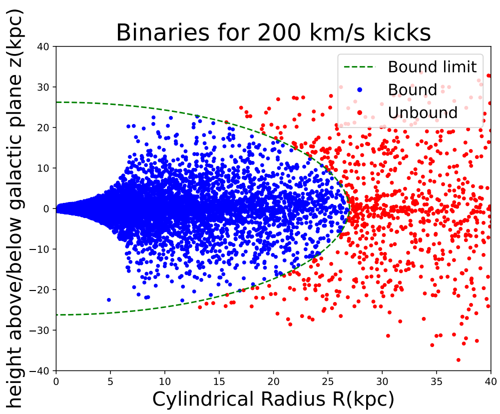
From the simulated orbits of our stellar objects, we can infer the retention fraction of stellar systems in terms of the kick magnitude. Here, retention fraction is defined as the number of systems remaining bound (retained) after the kick, divided by the total number of systems simulated. The definition of bound and unbound systems remain as introduced above. To obtain these results, we simply counted the number of bound systems for each simulated kick magnitude and plotted the results. For our initial simulation of kick magnitudes between 50 and 250 kms^{-1} with increments of 50 kms-1 the plot started to show a downward trend as expected. To have the opportunity to explore this trend, we decided to run our simulations up to 750 kms-1 using the same increments as before. Using the obtained results, we



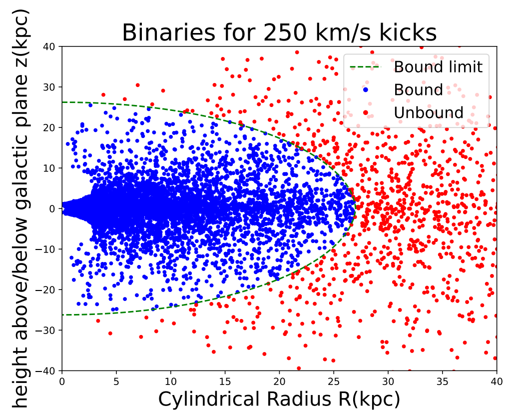
(a) 50 kms^{-1}



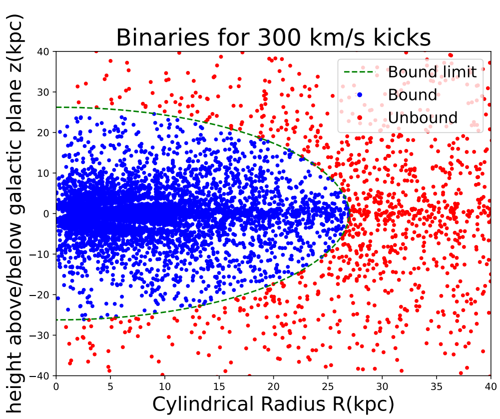
(b) 150 kms^{-1}



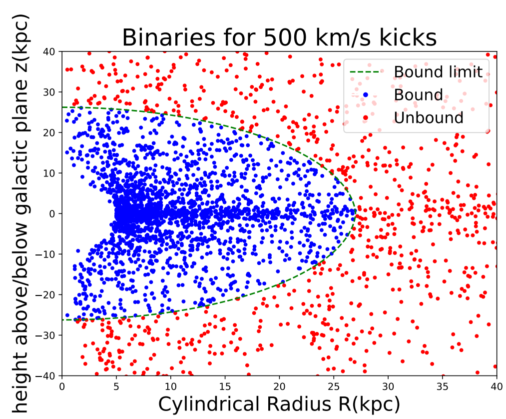
(c) 200 kms^{-1}



(d) 250 kms^{-1}



(e) 300 kms^{-1}



(f) 500 kms^{-1}

Figure 5: Plots illustrating the simulated systems, each point represents one system at the point where it was the least gravitationally bound. The systems in blue are the ones that remain bound and the ones that are red are unbound according to our definition. The green line shows the bound-unbound limit. Each of the six subplots shows the simulated systems for a fixed kick magnitude noted under them.

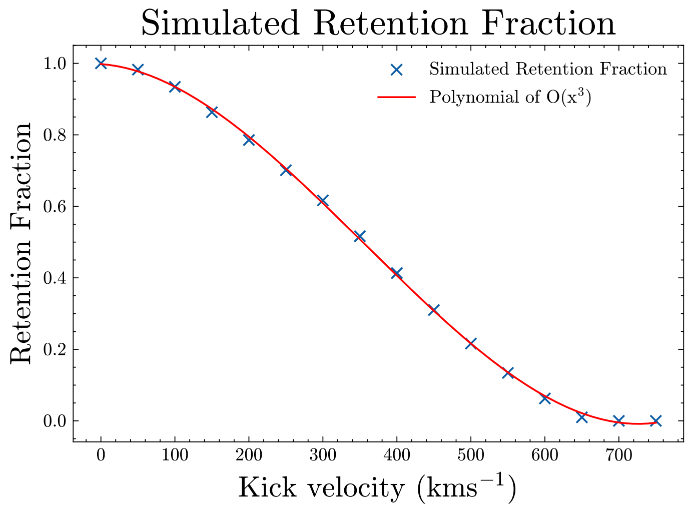
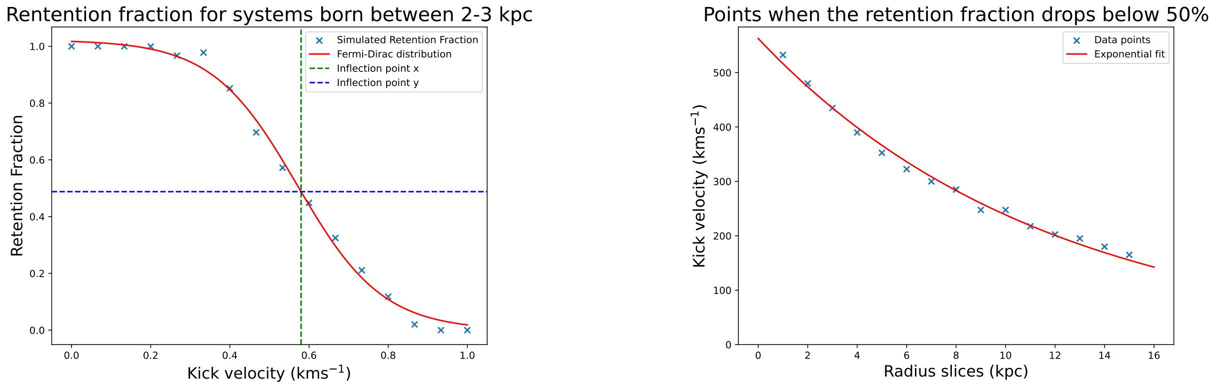


Figure 6: The retention fraction versus kick magnitude plot for 10,000 simulated systems, with a third order polynomial fit. A data point of unity was added for the kick value of zero, based on the feature of our simulation that all stellar systems are created bound. The uncertainties on the points arising from Poisson counting are not shown, since they are in the order of 0.01 and are not visible well.

tried fitting different functions to these. From visual inspection it is clear that a third order polynomial is the best fit out of the functions tried. The following parameters were determined for the polynomial: $4.96 \cdot 10^{-9}v^3 - 5.30 \cdot 10^{-6}v^2 - 1.54 \cdot 10^{-4}v + 1.00$, rounded to two significant figures. It is important to mention here that the Fermi-Dirac distribution fitted closely too, with a few deviations around the plateaus. We conduct additional analysis using this distribution to obtain partial retention fractions, demonstrated further below. The retention fraction versus kick magnitude plot for 10,000 simulated objects is presented in Figure 6.

After we found an overall trend for the retention fraction, we sliced our simulated galaxy into smaller segments containing still roughly a large number of stellar objects and produced retention fraction plots with best fit lines in these, an example of these is presented in Figure 7a. We sliced the Galaxy up into sixteen segments, with a 1 kpc range starting from the center until 15 kpc, with our last slice being everything beyond 15 kpc. For these partial retention fractions, the Fermi-Dirac distribution produced better fits than the third order polynomial, since the polynomial fit often overshot at the beginning of the steep drop in them. Thus, we fitted that for every segment. From these best fit curves, we determined the kick velocity where the retention fraction drops below 50% and noted down these values. From these values we created a plot of kick magnitudes required to drop the retention fraction below 50% versus the radial intervals. For this plot we excluded the last region incorporating everything from 15 kpc, since it is not a representative data point for this trend. We produced an exponential fit to this trend, with the following parameters: $713.25 \exp(-2.92\ (v - 6.86))$



(a) Partial retention fraction for systems between 2 and 3 kpc

(b) Kick velocity required at each radial segment to reach 50% retention

Figure 7: In the left figure, is an example for a partial retention fraction plot with the Fermi-Dirac distribution fitted to it. In the right figure, are the kick velocities required at each radial segment to drop the partial retention fraction below 50%. A decreasing exponential was fitted to these points.

rounded to two significant figure. This together with an example partial retention fraction plot is presented in Figure 7, while the partial retention fraction plots for different segments are included in Appendix C.

3.4 Intrinsic velocity distribution of pulsars after the natal kick

When investigating the kick velocities acquired by the systems, we are using the rest frame of the Galaxy as the reference frame. Previously, there have been many attempts at describing the velocity distribution of pulsars with respect to the Galactic rest frame. The most popular description for the observed kick magnitudes involve a Maxwell-Boltzmann distribution. Rather than outlining some physical phenomena, this is a computationally convenient choice and other distributions describe the data just as well (Verbunt et al., 2017). Some arrive at a bimodal Maxwellian distribution (Arzoumanian et al., 2002; Faucher-Giguère & Kaspi, 2006), which is two one-dimensional Maxwell distributions added together with relative scales, while others arrive at a single Maxwellian (Hobbs et al., 2005).

A recent study summarizing the previous models and the data used, together with newly available and more precise measurements of parallax and proper motions determines a bimodal Maxwellian distribution with parameters $\sigma_1 = 75$ kms^{-1} and $\sigma_2 = 316$ kms^{-1} and relative scales of 42% for the first one and 58% for the second one (Verbunt et al., 2017). Where, a Maxwellian distribution in velocity is given by the following equation:

$$p\left(v,\sigma\right) = \sqrt{\frac{2}{\pi}}\frac{v^2}{\sigma^3}\exp\left(-\frac{v^2}{2\sigma^2}\right). \tag{1}$$

Where v is the independent variable, in out case the velocity and σ is the parameter describing the distribution. From this, it is straight forward to show that a bimodal Maxwellian distribution with relative scales is as presented in Equation 2.

$$p\left(v, \sigma_1, \sigma_2\right) = f_1 \sqrt{\frac{2}{\pi}} \frac{v^2}{\sigma_1^3} \exp\left(-\frac{v^2}{2\sigma_2^2}\right) + f_2 \sqrt{\frac{2}{\pi}} \frac{v^2}{\sigma_2^3} \exp\left(-\frac{v^2}{2\sigma_2^2}\right) \tag{2}$$

Where σ_1 and σ_2 are the parameters of each of the Maxwellian distribution and f_1 and f_2 are the relative scales.

We obtained the observed velocity distribution of pulsars relative to the Galactic rest frame, and we want to infer the intrinsic kick velocity obtained by a pulsar when undergoing a supernova explosion. To do this, we assume that the intrinsic velocity distribution of pulsars can be approximately obtained from the observed velocity distribution and the retention fraction curve. The modified observed velocity plot with the retention fraction is presented in Figure 8.

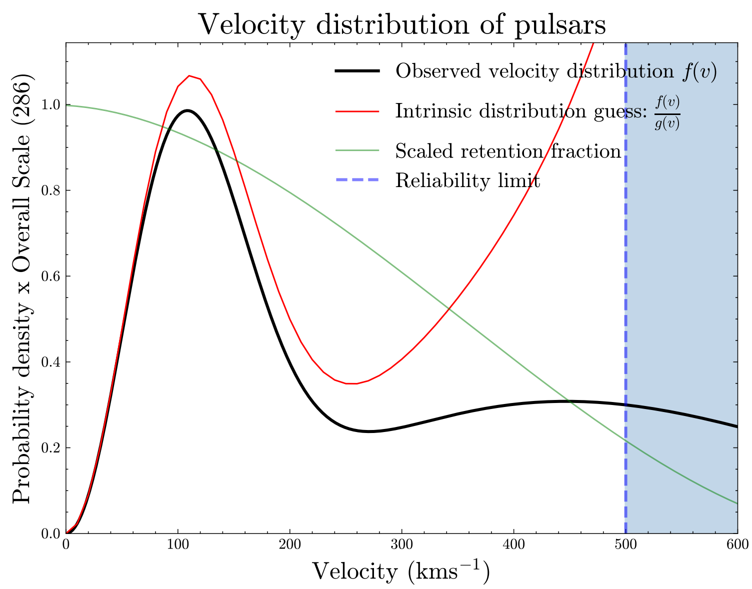


Figure 8: In this Figure our guess for the intrinsic kick velocity distribution of pulsars is presented. In black the original observed distribution, while in red the guessed curve for the intrinsic velocity distribution is shown. The blue dotted line represents the limit up until the retention fraction curve ($g(v)$) differs greatly from zero.

3.5 Exploring pulsar birth locations

To further showcase the prospects of the results obtained during the orbital integration, we decided to develop a process during which we can predict the possible radial birth location of pulsars based on their current proper motion and distance from the Solar System. The observed data that we used to create this method is presented in the Hobbs et al. (2005) study, henceforth, referred to as Hobbs2005. Specifically we were interested in the second data set presented there, which included pulsars with proper motions in both galactic longitude and latitude, together with distance measurements or estimates. The distance estimates were done using the dispersion measure or a Galactic electron density model outlined in Cordes & Lazio (2003).

Hobbs2005 presents a sky map of the pulsars with their proper motions indicated. This sky map uses the galactic longitude and latitude as coordinates and the proper motions are given in masyr^{-1} (milliarcsecond per year), meanwhile my simulations use a cylindrical polar coordinate system with an origin at the center of the Galaxy. To convert these variables we used the `astropy` package's unit conversion methods. Furthermore, to make sure that the dataset we obtained is consistent with the one in Hobbs2005, we recreated the first figure in their article, which is the sky map mentioned above.

To create projections on the sky, we used the `cartopy` package, specifically the Aitoff projection from it to be completely consistent with Hobbs2005. We realized that the galactic longitude points needed to be inverted in order to the get the correct view. This inversion holds no physical significance since it is due to the convention the `cartopy` package uses. Then from the proper motions we calculated the position on the sky of the pulsars 1 Myr ago and plotted their motion together with their origin point. The result is presented in Figure 9. We can see that the lines representing the motion inferred from the proper motion not being entirely smooth. This is due to the resolution of `cartopy` and it is not crucially important, since reproducing the plot was just a sanity check. Overall the location and the motions of the pulsars were consistent with Hobbs.

3.6 Radial distribution of possible birth location of pulsars from Hobbs2005

To determine the possible radial birth location of the pulsars form Hobbs2005 we first converted the galactic longitude and latitude to the cylindrical polar system we use. Then we plotted them in the R-z plane to make a visual comparison to our simulations. In Figure 10 a plot of the Hobbs2005 pulsars is presented with our bound-unbound definition. After converting the coordinates, we carry out the following procedure. We have to note here that since our potentials are axisymmetric, the azimuth angle of any of our simulated systems is completely arbitrary and by introducing some constant rotation we could have just as reasonable distribution of pulsars as we had before. Hence, the following search and matching algorithm works in two dimension, namely the cylindrical radius (R) and the distance from the Galactic plane (z).

For each pulsar from Hobbs2005 we iterate through all of our simulated data for each kick magnitude and we note the points that coincide with the Hobbs2005 pulsars ignoring the

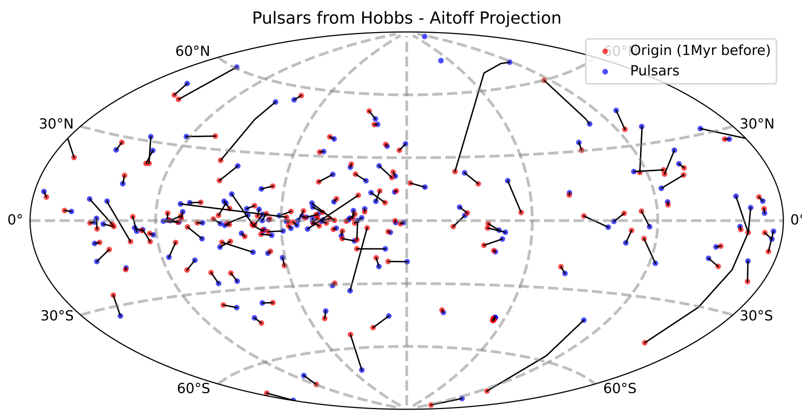


Figure 9: The Figure shows a sky maps of the pulsars obtained from Hobbs2005. The blue dots represent the current location of the pulsars, while the red ones are their locations 1 Myr ago inferred from their proper motion data. The `cartopy` package was used to create the Aitoff-projection, the sky coordinates are galactic longitude and latitude.

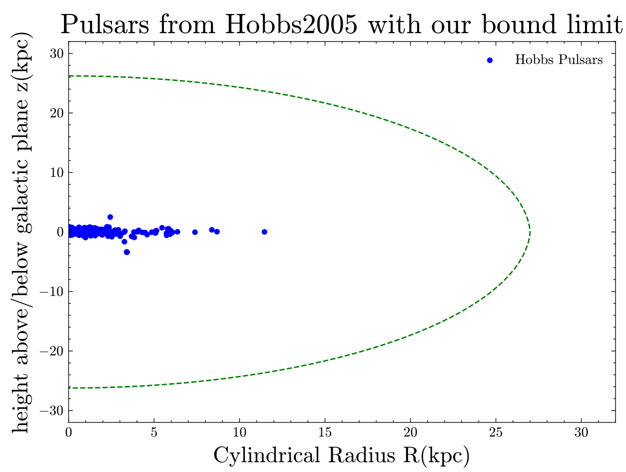
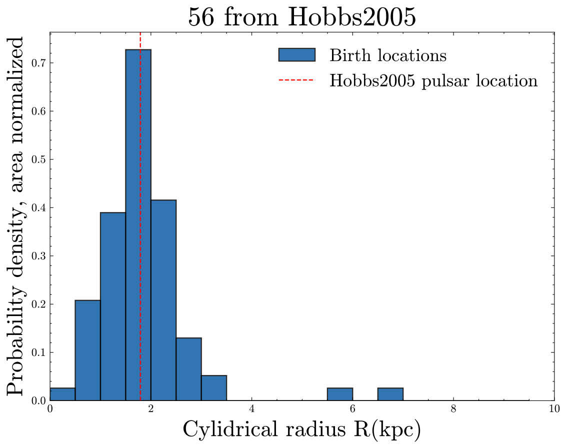
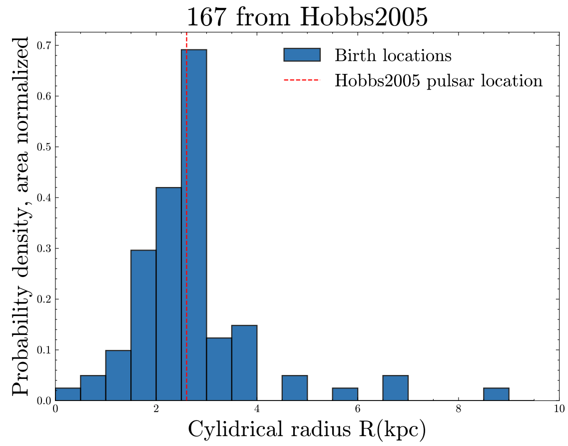


Figure 10: In the Figure the pulsars from Hobbs2005 are presented. The green line marks the definition of the bound limit that we made earlier.

azimuth angle here. By coincide we mean that the points reach within a circle of certain radius, we chose this radius to be 1.12 parsecs based on the following argument. If we draw a rectangle around the pulsars from Hobbs2005 in the R-z plane, it would approximately span from -3 to 3 kpc in the z direction and from 0to 10 kpc in the R direction. If we assume that for a each kick velocity all of our simulated data points ($1,000$ for each pulsar) would fall within this rectangle then on average each point would have an area of 1.25 pc^2 for itself. This area corresponds to a circle of radius 1.12 parsecs, so we define the condition when a simulated point coincides with a pulsar from Hobbs2005 to be when it is within of a circle of radius 1.12 parsecs of it in the R-z plane. If a pulsar during its simulated life goes through



(a) 76 matches



(b) 80 matches

Figure 11: In both Figures a distribution of the possible radial birth locations of a selected pulsar from Hobbs2005 is shown. In each, the observed cylindrical Radius of the pulsar is marked with a vertical red line. The observed location of the pulsars are ($R = 1.79$ kpc and $z = -0.01$ kpc) for the right one and ($R = 2.60$ kpc and $z = 0.01$ kpc) for the left one.

the selected point on multiple occasions it is recorded multiple times. After all the points have been recorded, we can select a pulsar from Hobbs2005 and plot the birth locations of the simulated pulsars going through the same point. Hence, creating a normalized histogram (distribution) of all the possible radial birth locations. In Figure 11 this predictive histogram is presented for the two pulsars with the most matches.

4 Discussion and Further research

4.1 The distribution of simulated systems with growing kick magnitudes

The plots in Figure 5 clearly outline the assumption that all the stellar system are born in the galactic plane ($z = 0$). At the low kick magnitude of 50 kms^{-1} the systems still resemble a straight line with a spread towards the higher radius ends. Since, here for each system the lowest gravitationally bound point is represented, the spread in the tail is due to the relatively large kick velocity in the vertical (z) direction. Furthermore, we can notice that at the unbound part the the spread starts to shrink again at large enough distances, because the radial kick needed to throw it that far in the radius is the significant contribution to the kick velocity. In the 150 kms^{-1} and 200 kms^{-1} plots the details of our potential starts to show. Especially, in the bound region, where at first we observe discontinuity in the spread. Following from the unbound orbits we see a linear spread up to a point in the bound regions, where it suddenly turns into a more rapid, exponential like spread, which is slower in the regions close to the center and speeds up as we go outwards. It is interesting to note that the empty regions around the center illustrate well that the simulated galaxy's mass density hence potential is the largest around the bulge, where significant amount of mass is within a small region. Thus, the velocity in the vertical direction to break away from this part increases rapidly, therefore filling the region slower. This feature again is well outlined on the 250 kms^{-1} plot, where only a small region around the center is inaccessible.

We can notice that the spread starts to disappear form the frame of the plot and the unbound systems starts to represent a background. This is well outlined in the 300 kms^{-1} and 500 kms^{-1} plots, where the only recognisable feature form our initial distribution remains the relative over-density of stars in the Galactic plane. Looking at all the systems in Figures 5e and 5f in Appendix A, we can see that there is still a trend for the linear spread on the former, but on the latter the systems tend to a shape defined by the large distance behaviour of the dark matter halo potential. The last thing to note, is the distribution of the bound systems near the galactic center. On the 300 kms^{-1} plot, we can see the entire galaxy we termed bound filled, with denser regions around the plane and less dense regions around the boundary. This illustrates that the 300 kms^{-1} magnitude is sufficient to overcome the resistance of high density regions. However, on the 500 kms^{-1} plot we see a cleared out galactic center which does not mean that there will be no stars there, since recall that the points are plotted at the coordinates where the systems are the least gravitationally bound. However, it shows that the additional 500 kms^{-1} moves the orbit quite significantly outwards. It is also interesting to note that the escape velocity of our simulation of the Milky Way is around 500 kms^{-1} at the distance of the Sun, so the bound stars we see on the plot originate mainly from the close vicinity of the galactic center. The escape velocity plot of our simulated galaxy is included in Appendix B.

4.2 Retention fraction and partial retention fractions

In Figure 6 we can observe a downward trend in the retention fraction versus kick velocity magnitude plot. This trend is well characterised by a third order polynomial fitted to the data, between 0 and 750 kms^{-1}. We observe that the last three data points are nearly or completely zero. This indicates that around 650 kms^{-1} all the systems will leave our limits of the galaxy. We can imagine that, by interpolating the bound systems plot for the 500 kms^{-1} kick shown in Figure 5f.

For the partial retention fraction plots the Fermi-Dirac distribution is a better fit. Since, at the low radius segments the retention fraction was remaining at unity even for large kick velocities. Also, more interestingly we can always observe the nearly or completely zero tail even at the lowest radial segments. This is due to the fact that the escape velocity from the our simulated galaxy is 700 kms^{-1} at the center and decreases from there outwards. Since, the kick magnitudes presented in the Figures are the additional velocity gain of the systems it is expected to see all of them escape above 650 kms^{-1}. Recall, that the systems are moving on circular trajectories around the center of the simulated galaxy, before receiving the kick, hence their overall velocity when receiving kicks above 650 kms^{-1}, will most likely be over the escape velocity. The only possible way that these systems can get retained if that a large portion of the isotropic kick acts directly opposite to their existing velocity.

The reason why we divided the simulated galaxy into sixteen segments in 1 kpc increments up to 15 kpc and then stopped afterwards and included everything else in the last slice is as follows. The sample sizes for the respective segments started to decrease rapidly following roughly the trend of the baryonic mass PDF presented in Figure 1 earlier, giving rise to large fluctuations within the data points. Hence, we decided to group the rest of the layers together. After fitting the Fermi-Dirac distribution for each of the segments, we found the velocity at with the respective partial retention fraction drops below 50% and noted the values down. Interesting to note, that the slice including everything beyond 15 kpc, does not follow the Fermi-Dirac distribution at all, but is well described by a third order polynomial.

The velocity magnitudes at which the retention fraction drops below 50% are presented in Figure 7b. We found that an exponential fit is sufficient to describe the downward trend. The exponential trend is reminiscent to the escape velocity curve of the simulated galaxy, just with an offset, due to the velocity difference. Also, since our definition of the bound-unbound limit it will not approach an almost straight line, but will decrease to zero. However, it does have a similar meaning, it gives an indication of the possible velocity increases the systems can handle. Furthermore, when looking at the 500 kms^{-1} plot of the evolved systems in Figure 5f, we can confidently say that those that remain bound come from the vicinity of the galactic center, Since systems at larger distances have exponentially less chance of remaining in the simulated galaxy after the kick.

An interesting aspect of the retention fraction fits is the fact that when fitting to the collective data of many radii, a third order polynomial trend is the best fit. However, when fitting to the radius segments the Fermi-Dirac distribution is used with success. It has some deviations in the mid-ranges around 8 kpc, where the high end tail is starting to grow and the fall of the retention fraction happens sooner, but there still are kick velocities with retention of a unity. It is interesting to note that the Fermi-Dirac distribution was developed for quantum statistics to describe the distribution of non-interacting identical fermions over energy states. The pulsars systems in our simulation are certainly non interactive, but otherwise do not have anything in common with quantum systems. Furthermore, the fact that for the collective behaviour a third order polynomial is the best fit suggest that there might exist a different functional description for the partial retention fractions which when combined yields the overall functional trend.

4.3 Intrinsic kick velocity distribution of pulsars

Our assumption was that, the observed velocity distribution ($f(v)$) is biased by the retention fractions ($g(v)$) at each velocity. For example, around a kick magnitude of 50 kms^{-1} the retention fraction is close to unity. Hence, we theoretically should observe close to all of the pulsars going with approximately 50 kms^{-1} relative to the rest frame of the galaxy. On the other hand, a kick velocity of around 360 kms^{-1} yields a retention fraction just below 50%, hence the observed pulsars with that velocity would only be half of what their initially were. To account for this, we assumed that the observed velocity distribution is the product of the intrinsic velocity distribution ($F(v)$) and the retention fraction versus velocity function. From this we can obtain the intrinsic velocity distribution:

$$F(v) = \frac{f(v)}{g(v)}.$$

Note, that this assumes that all the pulsars were born in very early stages of the galactic evolution and all of them had time to completely evolve and leave the galaxy. Furthermore, dividing the observed velocity distribution by the retention fraction function over-scales the large velocity end, due to the fact that the retention fraction reaches near zero levels around 600-650 kms^{-1}. Hence, a reliability limit is introduced to outline the regions where this scaling happens, this region is shaded blue in Figure 8. There are caveats with our attempt to describe the intrinsic velocity distribution of natal kicks, one of them is the high end tail getting a larger scaling outlined above. Other, is that the retention fraction plot, although including the underlying birth distribution of pulsars, is in overall a feature of our definitions in the simulations, such as the bound limit. A better way to infer an intrinsic distribution would be to select pulsars with measured parallaxes and proper motions and check our simulated systems that appear at the same position as the selected pulsars and from the matches create a velocity distribution that could have caused the pulsar to move there. We took a similar approach in Section 3.6, but we took it down a different path and analysed the possible radial birth distribution of selected pulsars.

4.4 Probabilistic radial birth distribution of pulsars

When discussing the birth distribution of pulsars a common assumption is to assume that all of the pulsars are initially born in the Galactic plane ($z = 0$) (Hobbs et al., 2005). This somewhat inspired our assumption when developing the simulations together with the simplification it provided. Due to this assumption, tracing back the pulsars to give a radial birth distribution of them is achievable from our simulations. In the plots presented in Figure 11, two pulsars from Hobbs2005 with the most matches appear. The histograms are normalized, which means that the area covered by them equals unity. We note that the observed location of these pulsars lays in the region with the highest probability, which is the case for most of the other pulsars even with much smaller match sizes. By visual examination the distribution seem to follow a Normal distribution. However, to verify this a functional fit would needed to be carried out. The distributions presented show that most of the stellar objects matching the pulsars come from their neighbourhood. It is interesting to note that both of the observed pulsars reside very near the galactic plane ($|z| = 0.01$ kpc), hence they are either very young and not travelled far or their kick velocity was concentrated to the Galactic plane. To further enhance this trace back the kick velocity and velocity at match for each object could be noted and compared it to the observed velocity of the pulsars. From this we could gain insight into the kick velocities needed to disturb the original systems in a way that they end up matching with an observed pulsar.

4.5 Further research

The first improvement, is that of running the simulation for the $10,000$ systems many times and noting the fluctuations it causes in the data points of the figures, such as the retention fraction versus kick velocity and partial retention fractions. From these fluctuations a systematic error could be calculated and presented. Secondly, extending the analysis done on the pulsars from Hobbs2005 with an additional velocity matching, between the observed pulsars and our simulated ones could be developed. Furthermore, newly discovered and more precisely measure pulsar data could be included to improve on our model for matching. Thirdly, this further improved matching model could be used to infer a more realistic intrinsic velocity distribution of the natal kicks.

5 Summary and Conclusion

We obtained orbital information for $10,000$ simulated systems using our galactic description. From this orbital information a polynomial for the total and an exponential trend for the partial retention fractions was observed. Using the polynomial trend for the total retention fraction and an observed kick velocity distribution an estimate of the intrinsic kick velocity distribution was inferred. Following this, we created a distribution of the possible birth locations of pulsars from Hobbs2005. Future methods are outlined for the improvement of the intrinsic distribution and the development and implication of the birth distributions. We conclude that our toy-model simulation based on the Milky Way with different natal kicks introduced to our systems, was successful in providing us with sufficient orbital information to observe trends in the properties of the evolved pulsars. The retention fraction trends predicted can be used as comparison for retention fraction estimates from studies using an observed kick velocity distribution for pulsars. A further improved pulsar matching algorithm together with an extended description incorporating satellite galaxies can be used to investigate pulsar contamination from nearby satellites. Also, the morphology of the simulated galaxy can be changed and the above trends can be investigated in different type of galaxies. The study can be further extended to compact binary systems with two natal kicks with the binaries simulated from stellar evolution codes.

References

Abadie J., Abbott B. P., Abbott R., Abernathy M., Accadia T., 2011, Physical Review D, 83

Abbott B. P., Abbott LIGO Scientific Collaboration Virgo Collaboration 2016, , 116, 061102

Arzoumanian Z., Chernoff D. F., Cordes J. M., 2002, , 568, 289

Bochanski J. J., Willman B., Caldwell N., Sanderson R., West A. A., Strader J., Brown W., 2014, , 790, L5

Bovy J., Rix H.-W., 2013, , 779, 115

Bray J. C., Eldridge J. J., 2018, Monthly Notices of the Royal Astronomical Society, 480, 5657–5672

Cordes J. M., Lazio T. J. W., 2003, NE2001.I. A New Model for the Galactic Distribution of Free Electrons and its Fluctuations (arXiv:astro-ph/0207156)

Deshpande A., Ramachandran R., Srinivasan G., 1995, Journal of Astrophysics and Astronomy, 16, 69

Duchêne G., Kraus A., 2013, Annual Review of Astronomy and Astrophysics, 51, 269–310

Faucher-Giguère C.-A., Kaspi V. M., 2006, , 643, 332

Hammer F., Puech M., Chemin L., Flores H., Lehnert M. D., 2007, The Astrophysical Journal, 662, 322–334

Hobbs G., Lorimer D. R., Lyne A. G., Kramer M., 2005, Monthly Notices of the Royal Astronomical Society, 360, 974–992

Igoshev A. P., 2020, Monthly Notices of the Royal Astronomical Society, 494, 3663–3674

Igoshev A., Verbunt F., 2018, in American Astronomical Society Meeting Abstracts #231. p. 132.03

Jánosdeák M., 2024, Master's thesis, University of Manchester

Kouveliotou C., Meegan C. A., Fishman G. J., Bhat N. P., Briggs M. S., Koshut T. M., Paciesas W. S., Pendleton G. N., 1993, , 413, L101

Kremer K., Breivik K., 2017, ApJ

Lamers H. J., M. Levesque E., 2017, Understanding Stellar Evolution. 2514-3433, IOP Publishing, doi:10.1088/978-0-7503-1278-3, https://dx.doi.org/10.1088/978-0-7503-1278-3

Li L.-X., Paczyński B., 1998, The Astrophysical Journal, 507, L59–L62

Lyne A., Manchester R., Taylor J., 1985, Monthly Notices of the Royal Astronomical Society, 213, 613

Manchester R. N., Hobbs G. B., Teoh A., Hobbs M., 2005, , 129, 1993

Mandhai S., Lamb G. P., Tanvir N. R., Bray J., Nixon C. J., Eyles-Ferris R. A. J., Levan A. J., Gompertz B. P., 2022, Monthly Notices of the Royal Astronomical Society, 514, 2716–2735

Miyamoto M., Nagai R., 1975, , 27, 533

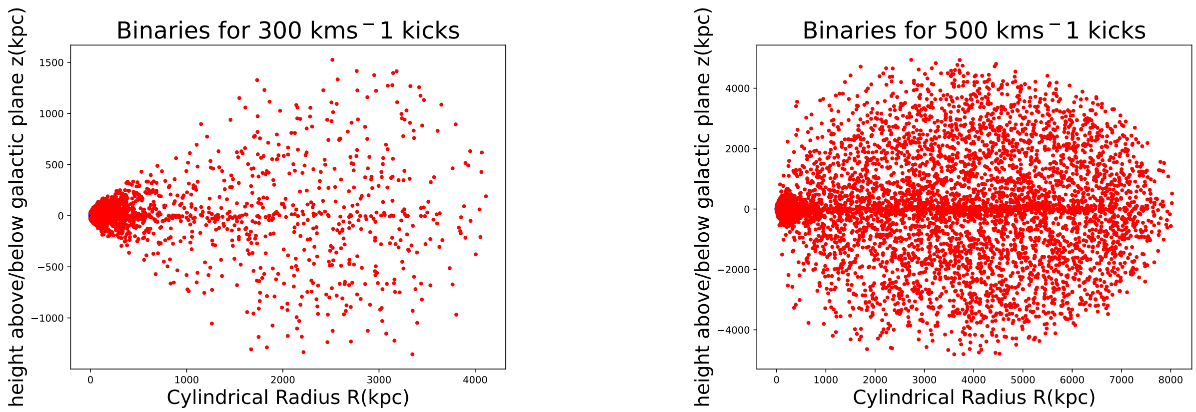
Navarro J. F., Frenk C. S., White S. D. M., 1996, , 462, 563

Tauris T. M., van den Heuvel E. P. J., 2006, in , Vol. 39, Compact stellar X-ray sources. pp 623–665, doi:10.48550/arXiv.astro-ph/0303456

Verbunt F., Igoshev A., Cator E., 2017, , 608, A57

Yang Y., et al., 2023, Gaia EDR3 proper motions, energies, angular momenta of Milky Way dwarf galaxies: a recent infall to the Milky Way halo (arXiv:2306.17208)

A All systems for 300 and 500 kms^{-1} kicks



(a) 300 kms^{-1} (b) 500 kms^{-1}

Figure 12: In both Figures all the simulated systems for the respective kick magnitudes of 300 and 500 kms^{-1} are presented.

B Escape velocity of the simulated galaxy

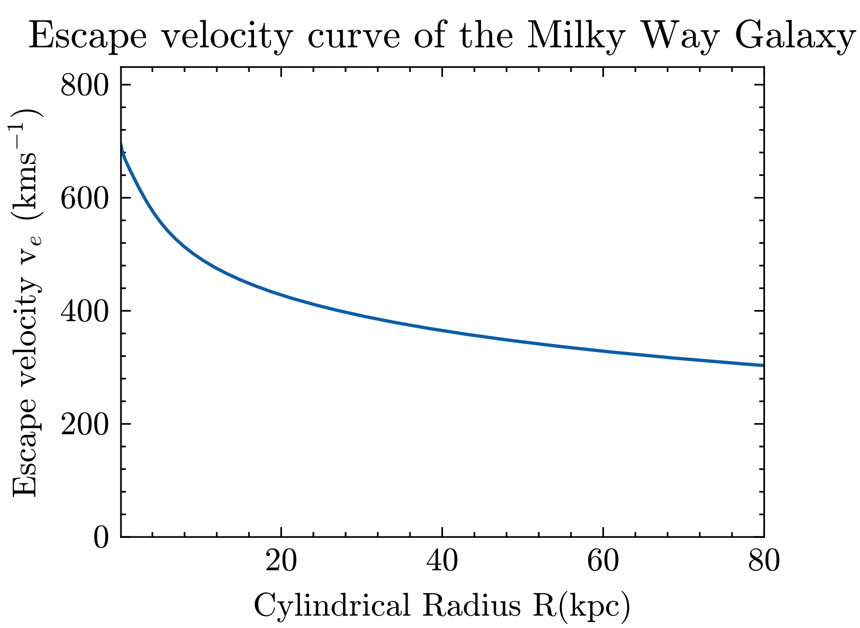
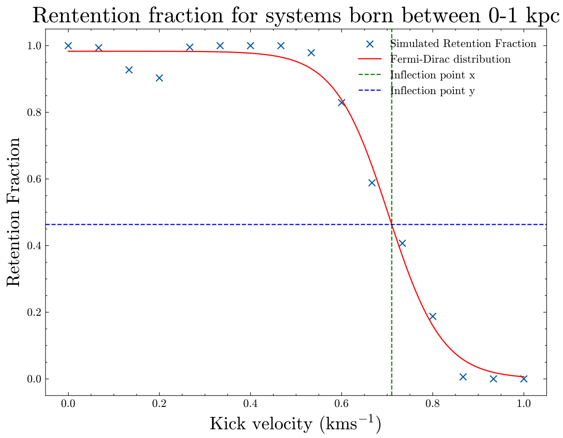
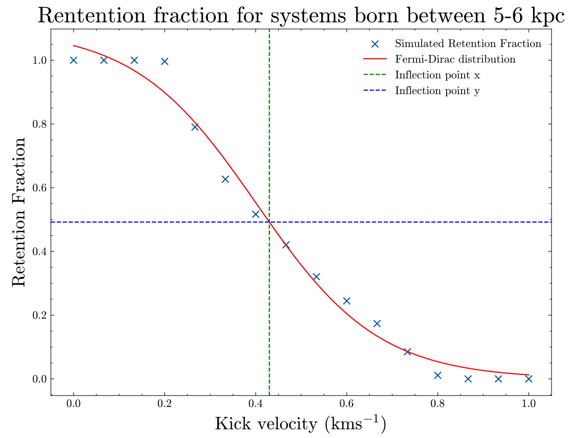


Figure 13: The escape velocity curve of our simulated galaxy versus the cylindrical polar radius is presented.

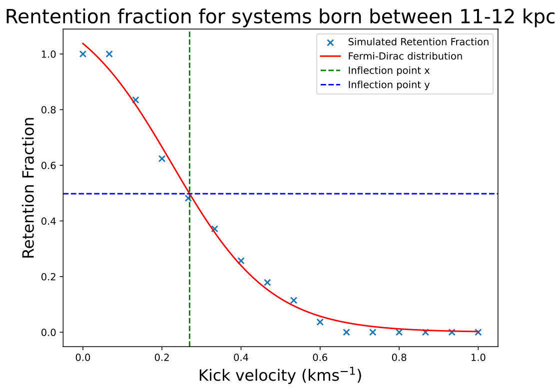
C Partial retention fractions



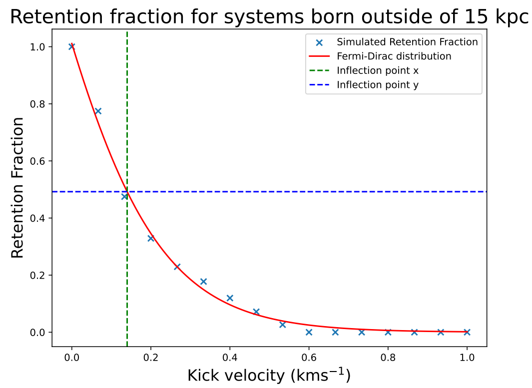
(a) 0-1 kpc



(b) 5-6 kpc



(c) 11-12 kpc



(d) 15+ kpc

Figure 14: In the plots above the partial retention fractions are presented for different sgements of our simulated galaxy.